

UBE2H expression and aneuploidy analyses

Purpose and scope

This document records the analysis procedures and dataset-specific decisions implemented in the accompanying R and R Markdown files.

The four datasets were analyzed differently because they provide different types of information:

Dataset	Data type	Analysis
GSE188644	Human bulk RNA-seq	Reversine-versus-DMSO differential-expression analysis
GSE313191	Mouse single-cell RNA-seq analyzed as pseudobulk	AA-versus-DD differential-expression analysis with lineage adjustment
GSE111672	Human PDAC single-cell RNA-seq	Per-cell chromosome-arm aneuploidy scoring derived from CopyKAT
GSE84465	Human GBM single-cell RNA-seq	Patient-specific per-cell chromosome-arm aneuploidy scoring derived from CopyKAT

Numerical results and biological conclusions are intentionally omitted. My purpose of this document is to describe what was done, which cells or samples were included, and how the reported statistics and figures were generated.

GSE188644: RPE-1 bulk RNA-seq

Dataset

GSE188644 contains bulk RNA-seq from human RPE-1 p53-knockout cells.

The full experiment contains 12 samples:

- three biological replicates;
- two treatments: DMSO and Reversine;
- two LMB2 overexpression states: Control and LMB2OE.

The treatment, LMB2OE status, and replicate information are encoded in the count-file names. Sample metadata are therefore parsed directly from the filenames rather than read from a separate metadata table.

Analysis subset

The differential-expression analysis is restricted to the six Control samples:

- three DMSO-treated Control samples;
- three Reversine-treated Control samples.

The six LMB2OE samples are excluded before DESeq2 model construction.

LMB2 overexpression was treated as a separate biological perturbation rather than as a technical covariate. Restricting the analysis to the non-overexpression background keeps the comparison focused on the effect of Reversine treatment without introducing the additional LMB2OE manipulation.

Count-matrix construction and gene filtering

Each input file contains a gene-symbol column and a raw-count column.

Before combining files, the notebook verifies that gene rows occur in the same order in every count file. The raw count columns are then combined into a single matrix.

Duplicate gene symbols are collapsed by summing their counts, producing one row per gene symbol.

After the six Control samples are selected, genes with fewer than 10 total counts across those six samples are removed before DESeq2 fitting.

Differential-expression model

DESeq2 is run on the six Control samples using:

```
design = ~ treatment
```

The reported contrast is:

```
c("treatment", "Reversine", "DMSO")
```

A positive log2 fold change therefore indicates higher expression in Reversine-treated cells relative to DMSO-treated cells.

Quality control and visualization

A variance-stabilizing transformation is calculated with:

```
vst(dds, blind = TRUE)
```

PCA is used as a sample-level diagnostic to inspect the relationship among the six retained samples.

The notebook also generates:

- a genome-wide volcano plot;
- a UBE2H boxplot using DESeq2-normalized counts;
- individual sample-level points overlaid on the UBE2H boxplot.

For the volcano plot, genes are classified as significant for display when:

- adjusted $p < 0.05$; and
- absolute log2 fold change is at least 1.

UBE2H is highlighted separately regardless of whether it meets those display thresholds.

Supplementary UBE2H tests

In addition to the DESeq2 analysis, the notebook performs Welch two-sample t -tests on:

1. DESeq2-normalized UBE2H counts; and
2. `log2(normalized UBE2H count + 1)`.

These t -tests are supplementary calculations and are separate from the DESeq2 differential-expression model.

GSE313191: ETiX single-cell RNA-seq analyzed as pseudobulk

Dataset

GSE313191 contains mouse ETiX embryoid single-cell RNA-seq data generated using the 10x Genomics platform.

Three targeted lineages are represented:

- ESC;
- iGata4;
- TSC.

Each lineage has three experimental conditions:

- **DD:** the lineage population is associated with the DMSO condition;
- **DA:** an intermediate or mixed DMSO/Reversine condition;
- **AA:** the lineage population is associated with the Reversine-treatment condition.

The complete dataset therefore contains nine samples: three lineages multiplied by three conditions.

Why a pseudobulk analysis was used

Previous attempts to generate a reliable per-cell aneuploidy classification using inferCNV and CopyKAT were not successful for this dataset.

In particular, the DD populations intended to serve as the reference condition showed substantial inferred CNV signal. This made it difficult to define a defensible cell-level diploid-versus-aneuploid classification.

The final workflow therefore does not assign individual cells an aneuploidy status. Instead, raw single-cell counts are summed within each sample, producing one pseudobulk expression profile per sample.

This approach changes the analysis question from a per-cell association between Ube2h and inferred aneuploidy to a sample-level comparison between the DD and AA experimental conditions.

The inferCNV and CopyKAT diagnostic workflows are not included in the supplied GSE313191 notebook. The final notebook contains the pseudobulk analysis only.

Why DA is excluded

DA is excluded before quality control and pseudobulk aggregation.

The final comparison is deliberately restricted to the two endpoint conditions:

- DD;
- AA.

DA is not modeled as an intermediate condition and does not contribute to the DESeq2 analysis.

After DA exclusion, six samples remain:

- ESC DD and AA;
- iGata4 DD and AA;
- TSC DD and AA.

Seurat-object construction

A metadata table is used to map each GEO accession to its corresponding sample name.

For each sample, the raw 10x files are read directly:

- matrix file;
- feature file;

- barcode file.

Feature names are taken from the second column of the feature table. Duplicate feature names are made unique using `make.unique()` before the Seurat object is created.

One Seurat object is generated per sample from the raw count matrix.

Cell-level quality control

Mitochondrial percentage is calculated using mouse mitochondrial gene names matching:

```
^mt-
```

The notebook first calculates three-median-absolute-deviation outlier flags for:

- detected features;
- total counts;
- mitochondrial percentage.

Detected features and total counts are log-transformed before the MAD calculation; mitochondrial percentage is not log-transformed.

These MAD-based flags are used for diagnostic summaries and pre-filtering violin plots. They are **not** used to remove cells.

The actual cell filtering uses fixed thresholds encoded in the notebook. A cell is retained when all of the following are true:

- `nFeature_RNA` is at least 500;
- `nFeature_RNA` is no greater than 5,000;
- `percent.mt` is below 25%;
- `nCount_RNA` is no greater than 50,000.

Equivalently, cells are removed when:

```
percent.mt >= 25 |  
nFeature_RNA < 500 |  
nFeature_RNA > 5000 |  
nCount_RNA > 50000
```

The notebook identifies these as thresholds taken from the original study.

The filter is applied using `subset()`. Post-filtering cell counts and violin plots are generated to verify that the filtering step was applied.

Pseudobulk aggregation

For each of the six retained DD or AA samples, raw counts are retrieved from the Seurat `counts` layer.

Counts are then summed across all QC-passed cells separately for every gene:

```
rowSums(raw_counts)
```

Each sample is therefore represented by one bulk-like gene-count vector.

Normalized single-cell expression values are not used to construct the pseudobulk matrix.

The resulting pseudobulk count matrix contains:

- genes as rows;
- six pseudobulk samples as columns.

Sample metadata

Lineage and condition are parsed from the sample names.

The condition factor is ordered with DD as the reference level:

```
condition = factor(condition, levels = c("DD", "AA"))
```

The three lineage levels are included as a separate factor.

Differential-expression model

DESeq2 is run using:

```
design = ~ lineage + condition
```

Including `lineage` accounts for baseline expression differences resulting from which ESC, iGata4, and TSC were treated.

The model estimates a shared AA-versus-DD condition effect after accounting for lineage. It does not fit a lineage-by-condition interaction.

Genes with fewer than 10 total counts across the six pseudobulk samples are removed before model fitting.

The reported contrast is:

```
c("condition", "AA", "DD")
```

A positive log2 fold change therefore indicates higher expression in AA relative to DD after adjustment for lineage.

Visualization

The notebook generates:

- a volcano plot of the lineage-adjusted AA-versus-DD results;
- a line plot showing DESeq2-normalized Ube2h counts in DD and AA for each lineage.

For the volcano plot, genes are classified as significant for display when:

- adjusted $p < 0.05$; and
- absolute log2 fold change is at least 1.

Ube2h is highlighted separately.

The displayed volcano-plot y-axis is limited to 0–75 so that an extreme genome-wide adjusted p -value does not compress the rest of the figure.

For the Ube2h plot, the DD and AA pseudobulk values are connected within each lineage. This plot is descriptive. Because there is one pseudobulk sample for each lineage-condition combination, it is not a separate within-lineage statistical test.

GSE111672: PDAC-A single-cell CopyKAT analysis

Dataset and analysis scope

GSE111672 contains human pancreatic ductal adenocarcinoma single-cell RNA-seq data generated using inDrop.

Two tumors, PDAC-A and PDAC-B, were profiled in the original study. The supplied analysis workflow is restricted to PDAC-A.

The PDAC-A expression matrix includes published cell-type labels for:

- ductal subtypes;
- Cancer clone A;
- Cancer clone B;
- immune populations;
- stromal populations;
- other normal pancreatic cell types.

Why PDAC-B was not carried forward

PDAC-B was excluded because a substantial fraction of ductal cells received CopyKAT aneuploid calls.

Ductal cells were intended to behave as a predominantly copy-number-normal comparison population relative to the two cancer clones. The unexpected CopyKAT behavior in PDAC-B reduced confidence in using its predicted diploid-cell signal as the baseline for the custom chromosome-arm scoring procedure.

PDAC-A showed a more interpretable relationship between the published cell identities and the CopyKAT calls and was therefore retained for the final analysis.

No PDAC-B CopyKAT object, notebook, or diagnostic output is included among the supplied files.

Input-matrix preprocessing

The input is the PDAC-A patient-specific inDrop filtered expression matrix.

Repeated column headers in the source matrix encode the published cell-type labels. The notebook generates unique cell identifiers of the form:

```
PDAC_A_cell_1
PDAC_A_cell_2
...
```

Gene symbols are stored in the first column of the input file.

Rows are retained when the gene symbol is:

- non-missing;
- non-empty;
- the first occurrence of that symbol.

Cells whose labels begin with `RBC` are removed before CopyKAT inference and before Seurat normalization.

CopyKAT reference and observation groups

Cell types are divided into two input groups using the cell-type label.

Cells whose labels contain either `Ductal` or `Cancer` are assigned to the observation group.

This includes:

- all Ductal subtypes;

- Cancer clone A;
- Cancer clone B.

All remaining cell types are assigned to the normal reference group. These include the immune, stromal, and other normal pancreatic populations present in the dataset.

The reference-cell identifiers are passed explicitly to CopyKAT using:

```
norm.cell.names = norm_cells
```

CopyKAT is therefore not left to identify its reference population automatically.

CopyKAT run

CopyKAT is run in the separate script:

```
GSE111672_run-copyKAT.R
```

The main GSE111672 analysis notebook does not rerun CopyKAT. It loads the saved CopyKAT `.rds` object.

The CopyKAT call uses:

```
copykat::copykat(
  rawmat = mat,
  id.type = "S",
  norm.cell.names = norm_cells,
  ngene.chr = 5,
  win.size = 25,
  KS.cut = 0.05,
  sam.name = "GSE111672_PDAC_A_inDrop",
  distance = "euclidean",
  genome = "hg20",
  n.cores = 12
)
```

The run script saves:

- the complete CopyKAT result object;
- a flat file containing cell-level CopyKAT predictions.

UBE2H expression normalization

A Seurat object is created from the same RBC-filtered raw count matrix.

Expression is normalized using:

```
NormalizeData(
  normalization.method = "LogNormalize",
  scale.factor = 10000
)
```

The normalized UBE2H value is retrieved from the Seurat `data` layer and stored as:

```
UBE2H_normalized
```

This Seurat-normalized value is used in the correlation tests, linear models, and expression plots.

CopyKAT bin-level signal

The CopyKAT `CNAmat` contains:

- bin annotations, including chromosome and genomic position;
- a CopyKAT signal for each genomic bin and cell.

The bin-level signal is used to derive chromosome-arm calls. The categorical CopyKAT prediction itself is not used as the per-cell aneuploidy score.

Derivation of the bin-signal threshold

A dataset-specific signal threshold is derived from cells that CopyKAT labels `diploid`.

All CopyKAT-predicted diploid cells are used for threshold estimation, regardless of whether they originally belonged to the reference or observation input group.

For those cells:

1. all finite CopyKAT bin-level signals are collected;
2. the absolute value of each signal is calculated;
3. the 90th percentile of the absolute signal distribution is used as the positive threshold;
4. the corresponding negative threshold is the same magnitude with the opposite sign.

In code:

```
bin_signal_threshold <- quantile(  
  abs(diploid_signal),  
  probs = 0.90,  
  na.rm = TRUE  
)
```

The threshold is therefore estimated from the background signal distribution of CopyKAT-predicted diploid cells in this specific CopyKAT run.

It is a threshold on the **bin-level CopyKAT signal**.

A histogram of the diploid-cell bin signals is plotted with the positive and negative thresholds marked.

Assignment of CopyKAT bins to chromosome arms

CopyKAT reports signal by genomic bin rather than by chromosome arm.

Each bin is assigned to a chromosome arm using:

```
data/arm_coords.csv
```

The assignment uses:

- chromosome identity;
- bin genomic position;
- chromosome-arm start and end coordinates.

A bin is assigned to an arm when its position is greater than or equal to the arm start and less than the arm end.

The code checks that no bin is assigned to more than one chromosome arm before continuing.

The short acrocentric p arms are excluded:

- 13p;
- 14p;
- 15p;
- 21p;
- 22p.

These arms do not contribute to the arm-call matrix or to the aneuploidy score.

Chromosome-arm calls

For each cell and chromosome arm, two quantities are calculated:

- the fraction of available bins with signal above the positive threshold;
- the fraction of available bins with signal below the negative threshold.

An arm is called:

- **Gain** when at least 80% of its bins exceed the positive threshold;
- **Loss** when at least 80% of its bins fall below the negative threshold;
- **Neutral** when neither criterion is met.

The corresponding numerical arm calls are:

Arm status	Numerical call
Loss	-1
Neutral	0
Gain	+1

Missing bin-level values are omitted when the gain and loss fractions are calculated.

The 80% requirement ensures that an arm call reflects a consistent signal across most of the assigned bins rather than a small number of isolated bins.

Aneuploidy score

For each cell, the full aneuploidy score is calculated as the number of chromosome arms called gained or lost.

A gained arm contributes one point, and a lost arm contributes one point.

In code:

```
aneuploidy_score <- colSums(abs(arm_call))
```

The score is therefore:

- a discrete integer count;
- a count of altered chromosome arms;

For example, a score of 5 indicates that five chromosome arms were called gained or lost.

The custom arm-count score is distinct from CopyKAT's categorical `diploid` or `aneuploid` prediction.

Chromosome 7q status

The call for chromosome arm 7q is extracted directly from the arm-call matrix.

Each cell is assigned one of three 7q states:

- Loss;
- Neutral;
- Gain.

UBE2H is located on 7q. Therefore, the analysis also calculates a second aneuploidy score after removing 7q from the arm-call matrix:

```
aneuploidy_score_no_chr7q
```

This score is constructed in exactly the same way as the full score, except that the 7q call does not contribute to the sum.

This prevents an alteration of the arm containing UBE2H from contributing directly to the predictor used to test the relationship between broader aneuploidy burden and UBE2H expression.

Analyses across all observation cells

An initial cell-level analysis is performed across all qualifying observation cells.

The observation group includes:

- all Ductal subtypes;
- Cancer clone A;
- Cancer clone B.

Cells are retained when they have:

- a CopyKAT-derived aneuploidy score;
- normalized UBE2H expression.

Within this observation-cell dataset, the notebook performs three related analyses.

UBE2H expression by 7q status

UBE2H expression is compared across the 7q Loss, Neutral, and Gain groups.

When at least two 7q-status groups are represented, pairwise Wilcoxon tests are performed with Benjamini-Hochberg adjustment.

Spearman correlation excluding 7q

A Spearman rank correlation is calculated between:

- normalized UBE2H expression;
- the per-cell aneuploidy score excluding 7q.

The correlation uses the original unbinned integer score for each cell:

```
cor.test(
  aneuploidy_score_no_chr7q,
  UBE2H_normalized,
  method = "spearman",
  exact = FALSE
)
```

Linear model adjusting for 7q status

A separate linear model is fitted:

```
UBE2H_normalized ~ aneuploidy_score_no_chr7q + chr7q_status
```

The model includes the 7q-excluded aneuploidy score and categorical 7q status as separate predictors.

This all-observation-cell model is distinct from the more restricted dataset used for the final boxplot.

Final cancer-cell, 7q-neutral analysis

The final displayed UBE2H-versus-aneuploidy analysis is restricted to:

- Cancer clone A cells;
- Cancer clone B cells;
- cells with a Neutral 7q arm call;
- cells with non-missing normalized UBE2H expression;
- cells with a non-missing 7q-excluded aneuploidy score.

Restricting the analysis to the two cancer clones prevents the Ductal populations from contributing to the reported cancer-cell correlation.

Restricting the analysis to 7q-neutral cells prevents variation in 7q copy-number status from explaining the displayed relationship.

Correlation and score-group visualization

Within the restricted cancer-cell, 7q-neutral subset, Spearman correlation is recalculated using:

- each cell's original 7q-excluded aneuploidy score;
- each cell's normalized UBE2H expression.

The score is **not** grouped before the correlation is calculated.

For visualization only, cells are assigned to the following score groups:

- 0–1;
- 2;
- 3 or greater.

The score groups are introduced because normalized single-cell UBE2H expression is zero-heavy and a simple cell-level scatterplot is difficult to interpret.

The plotted boxplots summarize UBE2H expression within the score groups, but the annotated Spearman coefficient and *p*-value are calculated from the original per-cell scores.

Boxplot outlier symbols are hidden using:

```
outlier.shape = NA
```

This changes only the display. Those cells remain in the boxplot statistics and in the Spearman correlation.

Diagnostic heatmaps

Two diagnostic heatmaps are generated.

Bin-level CopyKAT heatmap

The first heatmap displays the continuous CopyKAT bin-level signal by cell type.

Cells are ordered by cell type. Chromosome 23 is excluded from the displayed rows.

This heatmap is used to inspect the original CopyKAT signal before conversion to discrete chromosome-arm calls.

Chromosome-arm call heatmap

The second heatmap displays the discrete arm calls:

- blue for Loss;
- white for Neutral;
- red for Gain.

Cells are split by cell type and ordered within each cell type by the full chromosome-arm aneuploidy score.

A bar annotation shows the full arm-count score for each cell.

The heatmaps are diagnostic visualizations. They do not directly enter the correlation calculation.

GSE84465: GBM single-cell CopyKAT analysis

Dataset

GSE84465 contains human glioblastoma single-cell RNA-seq data generated using Smart-seq2.

Four patients are included:

- BT_S1;
- BT_S2;
- BT_S4;
- BT_S6.

The GEO series matrix provides cell-level metadata including:

- patient ID;
- cell type;
- Neoplastic/Regular status;

The published Neoplastic/Regular label is used to define the CopyKAT input groups.

Cells labeled `Regular` define the reference group.

Cells labeled `Neoplastic` define the observation group.

Raw-count preprocessing

The count matrix is read as a raw, non-negative integer matrix.

Genes with zero counts across every cell are removed.

The notebook verifies that:

- all matrix values are finite;
- all values are non-negative;
- all values are integers;
- the GEO metadata contain every cell present in the count matrix.

Why CopyKAT is run separately by patient

CopyKAT is run independently for each of the four patients.

Each patient has:

- a separate raw count matrix;
- a separate Regular reference population;
- a separate CopyKAT run;
- a separate diploid-cell signal threshold;
- a separate chromosome-arm call matrix.

CopyKAT bin-level signals are therefore not pooled across patients before threshold estimation or arm calling.

The patient-specific scores and metadata are combined only after CopyKAT inference and chromosome-arm scoring have been completed separately for each patient.

Cell filtering before CopyKAT

Before each patient-specific CopyKAT run, genes with zero counts in that patient are removed.

A light cell-quality filter is then applied.

Cells are retained for CopyKAT when they have:

- at least 200 detected genes;
- at least 500 total counts.

This filter is specifically applied to prevent near-empty cells from entering CopyKAT.

It is separate from the use of the full count matrix for Seurat normalization.

Patient-specific CopyKAT reference cells

Within each patient, that patient's Regular cells are identified after the CopyKAT input filter.

The Regular-cell identifiers are passed explicitly through:

```
norm.cell.names = norm_cells
```

CopyKAT is therefore not allowed to select its reference population automatically.

The code also verifies that every patient has:

- at least five reference cells;
- at least one observation cell.

CopyKAT run

The notebook can either:

- run CopyKAT when `params$run_copykat` is `true`; or
- load a saved combined CopyKAT result when `params$run_copykat` is `false`.

For each patient, CopyKAT is called using:

```
copykat::copykat(  
  rawmat = patient_counts,  
  id.type = "S",  
  ngene.chr = 5,  
  win.size = 25,  
  KS.cut = 0.1,  
  sam.name = patient,  
  distance = "euclidean",  
  norm.cell.names = norm_cells,  
  output.seg = "FALSE",  
  plot.genes = "TRUE",  
  n.cores = 4  
)
```

UBE2H expression normalization

A Seurat object is created from the full GSE84465 count matrix and associated cell metadata.

The Seurat object is constructed without an additional minimum-cell or minimum-feature filter.

Expression is normalized using:

```
NormalizeData(  
  normalization.method = "LogNormalize",  
  scale.factor = 10000  
)
```

UBE2H expression is retrieved from the Seurat `data` layer and stored as normalized expression.

Only cells that also occur in the patient-specific CopyKAT outputs contribute to downstream CopyKAT association analyses.

Patient-specific bin-signal thresholds

A separate bin-signal threshold is estimated for each patient.

Within a patient, all cells called `diploid` by CopyKAT are identified. This includes any CopyKAT-predicted diploid cells from either the original reference or observation group.

For each patient:

1. all finite CopyKAT bin-level signals from the predicted diploid cells are collected;
2. the absolute value of each signal is calculated;
3. the 90th percentile of the absolute values is used as that patient's positive threshold;
4. the negative threshold is the same magnitude with the opposite sign.

The analysis therefore produces four independently estimated thresholds rather than one threshold shared across the complete dataset.

Patient-faceted histograms display the diploid-cell signal distribution with the positive and negative thresholds marked.

Assignment of bins to chromosome arms

CopyKAT bins are assigned to chromosome arms independently for each patient.

Only bins on chromosomes 1–22 are considered.

Bin-to-arm assignment uses:

- chromosome;
- genomic position;
- arm start and end coordinates from `arm_coords.csv`.

The code checks that no bin is assigned to more than one arm.

The following short acrocentric p arms are excluded:

- 13p;
- 14p;
- 15p;
- 21p;
- 22p.

Chromosome-arm calls

Within each patient, arm calls are generated using that patient's signal threshold.

For every cell and arm:

- the fraction of bins above the positive threshold is calculated;
- the fraction of bins below the negative threshold is calculated.

An arm is called:

- **Gain** when at least 80% of its bins exceed the positive threshold;
- **Loss** when at least 80% of its bins fall below the negative threshold;
- **Neutral** otherwise.

Missing bin values are omitted when the fractions are calculated.

All arm calling is completed within each patient before patient-level results are combined.

Aneuploidy scores

The full per-cell aneuploidy score is the number of chromosome arms called gained or lost:

```
aneuploidy_score = colSums(abs(arm_call))
```

The score is a discrete integer count of altered chromosome arms.

A second score is calculated after removing 7q:

```
aneuploidy_score_no_7q
```

Only 7q is removed. Chromosome arm 7p remains included.

The 7q arm call is retained separately as:

- Loss;
- Neutral;
- Gain.

As in GSE111672, the full score and CopyKAT's categorical diploid/aneuploid prediction are different variables.

Combined cell-level metadata

After patient-specific arm calling, the four sets of cell-level results are combined.

The combined metadata include:

- patient;
- cell type;
- published Neoplastic/Regular status;
- CopyKAT prediction;
- Seurat-normalized UBE2H expression;
- full aneuploidy score;
- aneuploidy score excluding 7q;
- 7q arm status.

The combined CopyKAT predictions and cell-level metadata are also written as tab-delimited files under:

```
results/GSE84465/
```

Core 7q-neutral analysis dataset

One core analysis table is created and reused for the main GSE84465 association analyses.

Cells are retained when they are:

- in the observation group;
- therefore labeled Neoplastic in the published metadata;
- 7q-neutral;
- associated with one of the four included patients;
- non-missing for normalized UBE2H expression;
- non-missing for the 7q-excluded aneuploidy score.

This same dataset is used for:

- the pooled Spearman correlation;
- the pooled score-group boxplot;
- the patient-specific mean-and-SEM plot;
- the patient-adjusted linear model.

Pooled Spearman correlation

A Spearman rank correlation is calculated after pooling all eligible cells from the four patients.

The test uses:

- each cell's original, unbinned `aneuploidy_score_no_7q`;
- each cell's normalized UBE2H expression.

The test is implemented as:

```
cor.test(
  aneuploidy_score_no_7q,
  UBE2H_normalized,
  method = "spearman",
  exact = FALSE
)
```

The pooled Spearman test is cell-level and unadjusted for patient.

The number of cells and the number of represented patients are displayed with the pooled coefficient and *p*-value.

Pooled score-group boxplot

For visualization only, the original per-cell scores are assigned to the following groups:

- 0–1;
- 2–3;
- 4–5;
- 6 or greater.

These categories are used only for the x-axis of the boxplot.

The Spearman correlation is calculated before grouping.

The boxplot displays normalized UBE2H expression across the four score groups.

Boxplot outlier symbols are hidden using:

```
outlier.shape = NA
```

The corresponding cells remain in the boxplot calculations and in the Spearman correlation.

Patient-specific mean-and-SEM plot

The same score groups are used to create a patient-specific summary plot.

For every patient and score group, the notebook calculates:

- number of cells;
- mean normalized UBE2H expression;
- standard deviation;
- standard error of the mean.

The standard error is calculated as:

```
sem_expression = sd_expression / sqrt(n_cells)
```

The plot contains:

- one line per patient;
- one point for each available patient-by-score-group mean;

- error bars representing mean $\pm$ SEM.

(The pooled Spearman correlation is not calculated from the patient-specific means.)

The SEM error bars describe cell-to-cell variation within each patient and score group. They do not represent uncertainty across independent patients.

Patient-adjusted regression

A separate linear model is fitted on the same core Neoplastic, 7q-neutral cell set:

```
UBE2H_normalized ~ aneuploidy_score_no_7q + patient
```

Patient is included as a categorical covariate to account for baseline differences in normalized UBE2H expression among BT_S1, BT_S2, BT_S4, and BT_S6.

This regression is distinct from the unadjusted pooled Spearman test.

The coefficient for `aneuploidy_score_no_7q` represents the association between the unbinned score and normalized UBE2H expression after adjustment for patient-specific intercept differences.

Diagnostic heatmaps

Two diagnostic heatmaps are generated separately for each patient, giving eight heatmaps in total.

Bin-level CopyKAT heatmaps

One heatmap per patient displays the continuous CopyKAT bin-level signal.

Cells are assigned to four display groups:

- Regular reference;
- Neoplastic: diploid;
- Neoplastic: aneuploid;
- Not defined.

Cells are ordered within each group by the full arm-level aneuploidy score.

Chromosome-arm call heatmaps

A second heatmap per patient displays the discrete chromosome-arm calls:

- Loss;
- Neutral;
- Gain.

The same four display groups are used, and cells are again ordered within each group by the full aneuploidy score.

A bar annotation displays the score for each cell.

The heatmaps remain patient-specific because each patient's CopyKAT analysis and bin-signal threshold were generated independently. The bin-level signals are not combined into one cross-patient heatmap.

Key interpretation notes for the single-cell analyses

- The custom aneuploidy score counts altered chromosome arms.
- Gains and losses each contribute one point.

- The score is an unweighted integer count.
- The score is not weighted by arm length.
- The score is not weighted by the number of bins assigned to an arm.
- The score is not weighted by signal magnitude after an arm meets the calling criterion.
- CopyKAT's categorical diploid/aneuploid prediction and the custom chromosome-arm count are different variables.
- CopyKAT-predicted diploid cells are used to estimate the bin-signal threshold.
- In GSE111672, one threshold is calculated for the PDAC-A CopyKAT run.
- In GSE84465, one threshold is calculated independently for each patient.
- An arm is called altered only when at least 80% of its available bins cross the threshold in the same direction.
- The main UBE2H association score excludes 7q so that the chromosome arm containing UBE2H does not contribute to the predictor.
- Excluding 7q does not exclude 7p.
- The aneuploidy score is discrete. References to an unbinned or original score mean that the exact integer value for each cell is used.
- Score categories are used only for visualization.
- Spearman correlations are calculated using the original per-cell scores rather than the plotting categories.
- Hiding boxplot outlier symbols does not remove those cells from the statistical analyses.
- The GSE84465 pooled Spearman analysis is not patient-adjusted.
- The GSE84465 patient-adjusted linear model is a separate analysis.
- The patient-specific mean-and-SEM plot is descriptive and is not the basis of the pooled Spearman test.

Source analysis files

- `GSE188644_analysis.Rmd`
- `GSE313191_analysis.Rmd`
- `GSE111672_run-copyKAT.R`
- `GSE111672_analysis.Rmd`
- `GSE84465_analysis.Rmd`